# Hoa Huynh

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**LinkedIn:** linkedin.com/in/huynhthaihoa **Portfolio:** huynhthaihoa.github.io
**GitHub:** github.com/huynhthaihoa **LeetCode:** leetcode.com/u/huynhthaihoa

## ABOUT

Computer Vision Engineer with 7+ years in the field, including 5+ in industry developing and deploying real-time deep learning systems on edge devices for automotive safety and surveillance. Experienced in 2D/3D perception, inference optimization, and multi-sensor integration. Interested in Visual SLAM, 3D vision, and robotics.

## SKILLS

**Programming Languages**: C, C++, Python, C#, MATLAB

**Perception & Deep Learning**: Open3D, PCL, OpenCV, PyTorch, TensorFlow, Darknet

**Edge AI Deployment**: TI Edge AI/TIOVX (SK-TDA4VM, TDA4AEN), NVIDIA Jetson Nano/Xavier NX (TensorRT, DeepStream), Samsung Exynos V920, ONNX Runtime, TFLite

**Sensors & Calibration**: RGB/IR pinhole/fisheye, FLIR thermal, Intel RealSense, LUCID Helios2 ToF, Ouster LiDAR, ROS

**MLOps & Tooling**: Git, SVN, Docker, CMake, TensorBoard, DVC, W&B, MLflow, Jira, Confluence

**Databases**: Microsoft SQL Server, MySQL, SQLite, PostgreSQL, MongoDB

**Additional Languages & Frameworks**: Node.js, React, JavaScript, Java, LaTeX, WPF, Qt, FFmpeg

**Spoken Languages**: English (professional), Korean (elementary), Vietnamese (native)

## EXPERIENCE

**DeltaX** Seoul, South Korea
*Development Team Lead & AI Researcher* *June 2022 – July 2026*

- Led an R&D team from December 2023, driving the architecture, technical roadmap, and end-to-end delivery of an In-Cabin Monitoring System (ICMS) that aligned with Euro NCAP standards, while coordinating module development, milestone tracking, and client demonstrations across cross-functional teams.
- Architected a real-time edge AI pipeline using a single RGB/IR fisheye camera, C/C++ inference on an SoC (System-on-Chip), and a Node.js/React web visualization UI.
- Integrated 8 deep learning models (object detection, 2D/3D body keypoint estimation, facial landmark detection, gaze estimation, and image classification), achieving an average of 14 FPS on an 8-TOPS edge platform and 9 FPS on a 4-TOPS edge platform.
- Delivered 4 Proof-of-Concept (PoC) projects for 2 Korean automotive vendors within 3–9-month timelines: 42dot (Phase 1: January – June 2024; Phase 2: August – October 2024), Pyeong Hwa Automotive (Phase 1: May 2024 – January 2025; Phase 2: July – October 2025).
- Customized PyTorch models and optimized Python/C++ integration pipelines for SoC targets, deploying on TI SK-TDA4VM/TDA4AEN using Edge AI + TIOVX frameworks, and on Samsung Exynos V920 using an ONNX Runtime-based framework.
- Developed the 3D perception stack for worker localization, 3D display, and ADAS applications (Advanced Driver-Assistance Systems): monocular depth estimation (with pseudo-LiDAR conversion) trained under LiDAR/RGB-D supervision, semantic segmentation, visual SLAM, and real-time multi-camera stitching from fisheye and pinhole inputs.
- Designed a slicing-based preprocessing scheme derived from SAHI ("Slicing Aided Hyper Inference and Fine-tuning for Small Object Detection") in automotive body panel surface inspection to improve defect detection recall while maintaining real-time inference latency.
- Built the team's multi-sensor calibration, data collection, and analysis tooling (ROS, MATLAB, Python), and assembled real-world ADAS/in-cabin datasets across diverse subjects and lighting conditions.

**Smartdeep** Seoul, South Korea
*Research Engineer* *May 2021 – May 2022*

- Built a 16-channel CCTV surveillance system in C++ (OpenCV, FFmpeg) with a C# WPF front end, aligned with KISA Intelligent CCTV certification requirements for moving- and abnormal-object detection, recording, and event search.
- Owned dataset curation, training, and validation pipelines for Darknet YOLOv4 detection models.

**Ubay Solution** Seoul, South Korea
*AI Research Intern* *December 2020 – April 2021*

- Developed an OCR system for Busan Port to read truck license plates and container codes, benchmarking SSD, YOLOv4, and Faster R-CNN on a custom dataset expanded via augmentation and web crawling.
- Built a DeepStream + TensorRT pipeline on NVIDIA Jetson Nano/Xavier NX for real-time inference.

**ANDA Lab – Soongsil University** Seoul, South Korea
*Research Assistant* *September 2018 – August 2020*

- Diagnosed failure modes in a taillight-based vehicle localization system via experimental analysis in Python, OpenCV, and MATLAB, then designed the algorithms to address them: overlapping-LED detection for vehicular visible light communication (VLC), LED taillight detection for optical camera communication (OCC), and stereo-vision taillight matching and pairing for vehicle-to-vehicle positioning.
- First-authored 3 peer-reviewed papers on camera-based nighttime vehicle positioning, including IEEE Access (SJR Q1), MDPI Applied Sciences (SJR Q2), and J-KICS (SJR Q4).

**Global Cybersoft JSC** Ho Chi Minh City, Vietnam
*Software Engineer Trainee* *March 2017 – August 2017*

- Built C++ plugins for Japanese clients: MFC plugins for SCREEN Holdings' industrial printer system, and Adobe Illustrator SDK plugins (C++, JavaScript) for an advertising agency.

## EDUCATION

**Soongsil University** Seoul, South Korea
*Master of Engineering, Information and Communication Technology* *September 2018 – August 2020*

- **GPA:** 4.01/4.5
- **Thesis:** LED Taillight Matching Algorithm for Nighttime Vehicle Positioning *(written in English)*
- **Supervisor:** Professor Myungsik Yoo

**University of Information Technology – Vietnam National University** Ho Chi Minh City, Vietnam
*Bachelor of Engineering, Software Engineering* *August 2013 – June 2018*

- **GPA:** 8.6/10 (Salutatorian Honors)
- **Thesis:** Blood Vessel Segmentation on DICOM Images *(written in Vietnamese)*
- **Supervisor:** Thi-Vuong Pham, M.Sc.

## PUBLICATIONS

*(**J**: International Journal, **W**: International Workshop, **K**: Korean Journal/Conference)*

**K2**. Uddin S M Nadim et al. (including **Huynh Thai Hoa**), "A Railway Framework for Person (Worker) Localization based on Depth Estimation and GPS", *The Korean Institute of Defense Technology Fall Conference* (2024). Reference

**W1**. Jaime Spencer et al. (including **Huynh Thai Hoa**), "The Second Monocular Depth Estimation Challenge", *The IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops* (2023). Reference

**J2**. **Thai-Hoa Huynh**, Myungsik Yoo, "A Taillight Matching and Pairing Algorithm for Stereo-Vision-Based Nighttime Vehicle-to-Vehicle Positioning", *MDPI Applied Sciences* (2020). Reference

**K1**. **Thai-Hoa Huynh**, Myungsik Yoo, "Nighttime LED Taillight Detection Method with Optical Camera Communication System", *The Journal of Korean Institute of Communications and Information Sciences* (2020). Reference

**J1**. **Thai-Hoa Huynh**, Tuan-Anh Pham, Myungsik Yoo, "Detection Algorithm for Overlapping LEDs in Vehicular Visible Light Communication System", *IEEE Access* (2019). Reference

## SELECTED PROJECTS

**State Estimation Simulations**: Python simulations to compare robot pose estimation methods under noisy motion and sensor measurements, covering recursive filters (EKF, Invariant EKF, UKF), batch and incremental optimization (factor graphs, iSAM, bundle adjustment), and Lie-group math. GitHub

**Monocular Metric Depth Estimation**: PyTorch pipeline for monocular metric depth estimation, supporting pluggable encoders and decoders, benchmarking on NYUv2 and KITTI datasets, EdgeAI-TIDL-based quantization-aware training, and ONNX/TFLite model conversion for edge deployment. GitHub

**Notch-Based Point Cloud Registration**: C++ point-cloud registration aligning 3D scans to a reference model via notch-keypoint correspondence matching, for industrial inspection and quality verification. GitHub

**LiDAR-Based Ego Lane Detection**: Python signal-processing pipeline (ground extraction, intensity analysis, polynomial fitting) for interpretable ego-lane detection directly from LiDAR point clouds. GitHub

*(Full project list available on portfolio: huynhthaihoa.github.io/#projects)*

## SELECTED CERTIFICATIONS

**Coursera**: Introduction to Machine Learning in Production.

**Hugging Face**: Fine-tuning Language Models, AI Agents Fundamentals, Fundamentals of Group Relative Policy Optimization.

*(Full certification list available on portfolio: huynhthaihoa.github.io/#certifications)*